

# Pairs of Random Variables

## Functions of two random variables

► **continuous case:** the CDF of  $W = g(X, Y)$  is obtained via

$$F_W(w) = P(W \leq w) = \iint_{g(x,y) \leq w} f_{X,Y}(x, y) \, dx dy$$

▷ in particular, the CDF of  $W = \max(X, Y)$  is

$$F_W(w) = \int_{-\infty}^w \int_{-\infty}^w f_{X,Y}(x, y) \, dx dy$$

and the CDF of  $W = \min(X, Y)$

$$F_W(w) = 1 - \int_w^\infty \int_w^\infty f_{X,Y}(x, y) \, dx dy$$

• the expected value is

$$E(W) = \int_{-\infty}^\infty \int_{-\infty}^\infty g(x, y) \cdot f_{X,Y}(x, y) \, dx dy$$

► **discrete case:** for a pair of discrete random variables it is more effective to work with the PMF of the random variable  $W = g(X, Y)$

$$P_W(w) = \sum_{(x,y): g(x,y)=w} P_{X,Y}(x, y)$$

• the expected value of  $W = g(X, Y)$  will be

$$E(W) = \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) \cdot P_{X,Y}(x, y)$$

## Variance and expected value

• the expected value is linear but not multiplicative

$$E[aX + bY] = a \cdot E[X] + b \cdot E[Y]$$

because only for **independent** random variables  $X, Y$  one has

$$E[X \cdot Y] = E[X] \cdot E[Y]$$

thus we define the ► **covariance** to measure how much  $E$  fails to be multiplicative

$$\text{cov}(X, Y) = E[X \cdot Y] - E[X] \cdot E[Y]$$

• the variance is not even linear, since again only for **independent** random variables

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

• for general random variables we use

$$\text{var}(aX + bY) = a^2 \cdot \text{var}(X) + b^2 \cdot \text{var}(Y) + 2ab \cdot \text{cov}(X, Y)$$

► **pairs of continuous random variables:** the joint probability density function of the continuous random variables  $X$  and  $Y$  is a function  $f_{X,Y}(x, y)$  with the properties

$$f_{X,Y}(x, y) \geq 0 \quad \text{and} \quad \int_{-\infty}^\infty \int_{-\infty}^\infty f_{X,Y}(x, y) \, dx dy = 1$$

• the joint cumulative distribution function is defined by

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) \, du dv$$

and it can be used for evaluating probabilities

$$P(a \leq X \leq b, c \leq Y \leq d) = F_{X,Y}(b, d) - F_{X,Y}(b, c) - F_{X,Y}(a, d) + F_{X,Y}(a, c)$$

• also, the probability of an event  $A$  is obtained by double integration over the region described by  $A$

$$P(A) = \iint_A f_{X,Y}(x, y) \, dx dy$$

• the marginal probability density functions can always be obtained from the joint PDF

$$f_X(x) = \int_{-\infty}^\infty f_{X,Y}(x, y) \, dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^\infty f_{X,Y}(x, y) \, dx$$

• but only for **independent random variables** one can recover the joint PDF from the marginal PDF's

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$$

► **pairs of discrete random variables:** change integrals with possibly infinite sums (numerical series) in the above statements, in order to obtain the similar statements for a pair  $(X, Y)$  when both are discrete

• keeping in mind that  $\{X = x, Y = y\}$  is an event in an experiment, the sample space becomes now

$$S_{X,Y} = \{(x, y) : P_{X,Y}(x, y) > 0\}$$

• for discrete random variables  $X, Y$  the characteristic property of the joint PMF becomes

$$\sum_{x \in S_X} \sum_{y \in S_Y} P_{X,Y}(x, y) = 1$$

• the probability of the event  $A$  is

$$P(A) = \sum_{(x,y) \in A} P_{X,Y}(x, y)$$

• the marginal probability mass functions are

$$P_X(x) = \sum_{y \in S_Y} P_{X,Y}(x, y), \quad P_Y(y) = \sum_{x \in S_X} P_{X,Y}(x, y)$$

## Pairs of Random Variables

In some situations, for example the ► Problem of Meeting, one needs two random variables  $X$  and  $Y$  for creating a mathematical model. The random variable  $X$  will store the time when the first person arrives in the restaurant and  $Y$  will store the arrival time of the second person.

• the correlation coefficient  $\rho_{X,Y}$  is a measure of linear ► **correlation** between two sets of data, but beware that ► **correlation doesn't imply causation**

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X) \cdot \text{var}(Y)}} = \frac{\text{cov}(X, Y)}{\sigma(X) \cdot \sigma(Y)}$$

where  $\sigma(X), \sigma(Y)$  are the standard deviations of the two random variables

## Conditioning by an event

• all the formulae are very similar to the case of a single random variable: for any event  $B$ , a region of the  $X, Y$  plane with  $P(B) > 0$ , the conditional joint probability mass function of  $X, Y$  given  $B$  is

$$P_{X,Y|B}(x, y) = \begin{cases} \frac{P_{X,Y}(x, y)}{P(B)}, & \text{if } (x, y) \in B \\ 0, & \text{otherwise} \end{cases}$$

• the conditional joint probability density function of  $X$  and  $Y$  is

$$f_{X,Y|B}(x, y) = \begin{cases} \frac{f_{X,Y}(x, y)}{P(B)}, & \text{if } (x, y) \in B \\ 0, & \text{otherwise} \end{cases}$$

## Conditioning by a random variable

• for any event  $Y = y$  such that  $P_Y(y) > 0$ , the conditional PMF is

$$P_{X|Y}(x|y) = P(X = x | Y = y)$$

• one can observe that

$$P_{X|Y}(x|y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{P_{X,Y}(x, y)}{P_Y(y)}$$

thus for a pair of random variables  $X, Y$  one has the general rule

$$P(X = x, Y = y) = P_{X|Y}(x|y) \cdot P_Y(y) = P_{Y|X}(y|x) \cdot P_X(x)$$

• for a continuous random variable the conditioning PDF of  $X$  given  $Y = y, f_Y(y) > 0$  is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

• there are similar formulae for the conditional expected value

$$E(X|Y = y) = \sum_{x \in S_X} x \cdot P_{X|Y}(x|y)$$

$$E(g(X, Y)|Y = y) = \sum_{x \in S_X} g(x, y) \cdot P_{X|Y}(x|y)$$

and in the continuous case

$$E(X|Y = y) = \int_{-\infty}^\infty x \cdot f_{X|Y}(x|y) \, dx$$

$$E(g(X, Y)|Y = y) = \int_{-\infty}^\infty g(x, y) \cdot f_{X|Y}(x|y) \, dx$$